Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 1 of 5 · Question 8 part whole · 6 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

Question number			
8 (a)	**Advantages embryonic stem cells:** • generate most cell types; • can maintain and grow for long lengths of time in culture; • useful for drug development studies/Parkinson's disease; **Disadvantages of embryonic stem cells:** • can cause tumours/cancer; • many embryos destroyed; • some people think it is unethical/killing/murder; • possibility of rejection if used in transplants **Advantages of adult stem cells:** • can be reprogrammed; • less likely rejected if used in transplants; • taken from the patient itself; **Disadvantages of adult stem cells:** • limited ability to differentiate/develop into different types of cell; • cannot be grown for long periods of time in culture; • usually a very small number found in body tissues; • can only grow small numbers;	Three marks from embryonic stem cell section and three marks from adult stem cell section Allow a valid named ethic	6
			Total 6

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 1 of 5 · Question 8 part whole · 6 marks

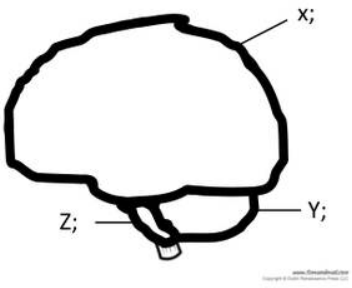
2022 JUN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 2 of 5 · Question 7 part c · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

7 (a)		Letters can be anywhere within outline shown. Allow labels in place of than letters.	3
(b) (i)	Motor;		1
(ii)	Any one from the following: (sensory neurone): longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord (motor neurone) shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord	Answer must be comparative and state which type of neurone is being referred to	1
(iii)	• chemicals/neurotransmitters; • diffuse; • across a synapse; • bind to receptors (on post-synaptic membrane);		3
(c)	• stem cells are undifferentiated/unspecialised/can differentiate; • could be made to form nerve cells; • to replace damaged cells in brain;		1 1 1

Total question 7 = 11 marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 2 of 5 · Question 7 part c · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 3 of 5 · Question 6 part c · 1 mark

2023 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

6 (a)	• fertilisation; • egg and sperm fuse/gametes/join;		2
(b)	Any four from • mitosis; • DNA is replicated; • chromosomes move to the centre of the cell; • chromosomes are separated to opposite poles; • cell divides;	Allow one mark for prophase, metaphase, anaphase, telophase in the correct order Allow zygote/embyro divides	4
(c)	**D;** (the cells are able to differentiate into any type of body cell)		1
(d)	• (corpus luteum) produces progesterone; • to maintain uterus lining;		2
(e)	Any two from: • umbilical cord established; • transports nutrients/named nutrients/oxygen (to fetus/embryo); • removes waste products/named waste product (from fetus/embryo);		2
(f)	development of male sexual characteristics/genitalia;		1
(g)	**C;** (release of oxytocin from the pituitary gland)		1

Total for Question 6 = 13 marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 4 of 5 · Question 6 part a.i · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
6	(a)	(i)	Three from: • the use of embryonic stem cells is not needed/embryos not destroyed; • easily available source of cells/easy to obtain/no need to find a donor; • same cell/tissue/blood type of the patient/same antigens; • reduced chance of rejection/not recognised as foreign;		3
		(ii)	Two from: • unlicensed clinics could be used; • stem cells could develop into tumours/cancer; • risk of rejection (if the cells are not the recipients); • increased risk of infection;		2
	(b)	(i)	860 000 and 480 000; 380 000;	Ecf allow one mark for incorrectly derived figures from graph but correct answer from these figures	2
		(ii)	**D** structure Z;		1
	(c)		Three from: • cut out desired/beta-carotene/gene/gene for vitamin A; • using/cut with restriction enzymes; • insert into plasmid; • insert into/infect rice genome; • to produce golden rice;		3

Total 11 marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 4 of 5 · Question 6 part a.i · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 5 of 5 · Question 6 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
6	(a)	(i)	Three from: • the use of embryonic stem cells is not needed/embryos not destroyed; • easily available source of cells/easy to obtain/no need to find a donor; • same cell/tissue/blood type of the patient/same antigens; • reduced chance of rejection/not recognised as foreign;		3
		(ii)	Two from: • unlicensed clinics could be used; • stem cells could develop into tumours/cancer; • risk of rejection (if the cells are not the recipients); • increased risk of infection;		2
	(b)	(i)	860 000 and 480 000; 380 000;	Ecf allow one mark for incorrectly derived figures from graph but correct answer from these figures	2
		(ii)	**D** structure Z;		1
	(c)		Three from: • cut out desired/beta-carotene/gene/gene for vitamin A; • using/cut with restriction enzymes; • insert into plasmid; • insert into/infect rice genome; • to produce golden rice;		3

Total 11 marks

Type MS · Explain and evaluate uses of embryonic and adult stem cells

Answer 5 of 5 · Question 6 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks